A Two-Layer Synthetic Data Generator for Studying Student Retention Early-Warning Systems in Philippine Higher Education

Matthew C. Batitay

Eskwelabs Innovation Fellowship, Data Innovation Track

Abstract. Administrative records at Philippine higher-education institutions capture what a student does (grades, attendance, financial-account status) but rarely why a student leaves. Published four-year cumulative attrition figures also disagree: a single CHED regional table reports two consecutive graduating cohorts whose regional rates differ by about nineteen percentage points on average, and the national figures in circulation, roughly three and roughly four in ten students who begin a degree, do not appear to trace to the same cohort or source. Section 10 traces them and states which cohort the model calibrates to, and why. Whichever figure is used, no public dataset contains the individual-level mechanisms behind these numbers. This paper specifies a synthetic data model that reproduces this gap deliberately. A latent state (a student's structural vulnerability and exposure to acute shocks) drives both the observed institutional record and the eventual dropout, yet stays unobservable to any downstream analyst. A confounded intervention-assignment mechanism, motivated by the documented tendency of early-warning systems to base support on the same signals that predict the outcome, means a support program's effectiveness cannot be read off a naive comparison. Every generative parameter is tagged as locked, calibrated against cited Philippine statistics, or explicitly assumed. The model's aggregate behavior is calibrated against national and regional attrition figures reported by the House of Representatives' Congressional Policy and Budget Research Department, the Second Congressional Commission on Education, and the Philippine Institute for Development Studies, using the calibration loop of Section 11, which that section distinguishes from a limited, pre-specified held-out check against a second graduating cohort. A pilot instantiation (Appendix A) calibrates to the tier targets and reproduces the intended confounding, but only estimators that borrow strength across signal cells recover the true effect, and its held-out check lands marginally outside the pre-specified band.

1 Introduction

National figures on Philippine higher-education attrition describe a large problem but disagree on its size. Reported four-year cumulative attrition lies between roughly three and four in ten students who begin a bachelor's degree, depending on the figure and the graduating cohort it describes (Section 10 traces the figures to their cohorts and states which one the model calibrates to and why). National averages also conceal wide regional spread: in CHED's regional table, cumulative attrition ranges from 36.3% to 93.4% across regions for the Class of 2022 and from 8.0% to 38.4% for the Class of 2023 (BARMM unreported) (CPBRD, 2025). These figures are

collected by region, institution, or cohort, and by design cannot be collected for the individual student whose circumstances led to a specific outcome.

Institutional records therefore miss much of what determines a student's trajectory. A registrar's database holds grades, attendance flags, and tuition-account status. It does not hold, and generally has no lawful or practical way to hold, a reliable measure of a student's financial precarity, household stability, or whether an acute event (a medical emergency, family crisis, or unexpected expense) has just made continued enrollment impossible. Qualitative research on Philippine dropout consistently identifies financial strain, family and employment pressures, and weak institutional support as dominant themes in students' own accounts (Cahilog et al., 2025; Naz, 2024), yet these factors are almost never columns in an administrative dataset.

Because the individual-level data needed to study this gap does not exist publicly, and for privacy reasons should not, any retention early-warning system faces a basic evidentiary problem: the ground truth for checking predictions is never observed, even by the institution generating the data. This paper instead specifies a synthetic data model that treats the gap between recorded and true state as the central object being modeled, not as noise to engineer around.

The model separates each student's situation into two layers. An observed layer (grades, attendance, financial-hold status, and whether the student received institutional support) is exportable, analyzable, and structurally similar to a real student-information system. A latent layer (a fixed baseline vulnerability and a per-term probability of an acute shock) drives the observed layer and the eventual dropout but is never exported and is invisible to any analysis of the dataset. This deliberate gap reproduces the epistemic position of a Student Success Analyst working from institutional records alone. (The title is used increasingly in Philippine and international higher-education administration for the staff role responsible for retention analytics and intervention targeting, though it is not yet a single standardized job title; here it stands in for whichever office owns that responsibility.)

The contribution is the data model itself: the entity structure, the latent-to-observed generative logic, the calibration targets drawn from published Philippine statistics, and the validation checks any implementation must satisfy. The generator's source code and the resulting dataset are separate deliverables built on this specification. Section 2 reviews the motivating literature, Section 3 states the design principles, and Section 4 describes the entity inventory and pipeline. Sections 5–9 detail the five phases. Section 10 specifies the calibration parameters and provenance tags, Section 11 the validation measures and benchmarks, and Section 12 two engineered edge cases. Sections 13, 14, and 15 cover ethical considerations, limitations, and conclusions. Appendix A gives a reference instantiation and pilot check, and Appendix B the field-level schema and reproducibility policy.

2 Related Work and Theoretical Foundation

2.1 Classical Attrition Theory

Tinto's model frames dropout as the end point of insufficient academic and social integration and remains the most widely cited theoretical account of college attrition (Tinto, 1975). Bean and Metzner extend it to non-traditional students, arguing that external factors (finances, employment, family responsibility) dominate campus integration for students who are older, commuting, or working while enrolled (Bean and Metzner, 1985). Both converge on the claim motivating this paper: the recordable events preceding a dropout (a missed class, a failed course, a lapsed payment) are downstream symptoms of a state recorded nowhere. The two-layer structure of Section 6 operationalizes this claim.

2.2 Machine-Learning Dropout Prediction and the Actionability Gap

A large and growing literature applies machine-learning classifiers to institutional and learning-management-system data to predict dropout, frequently reporting high accuracy. A 2026 systematic review of 75 quality-assessed studies under PRISMA guidelines finds limited evidence of how these models translate into empirically validated interventions in real institutions, a gap it terms the actionability gap (Lopez-Muñoz et al., 2026). Because a model that predicts risk accurately is distinct from a program that demonstrably reduces it, this paper's data model treats prediction and intervention effectiveness as two questions with separate, independently checkable answers.

2.3 Confounded Treatment Assignment in Educational Interventions

The actionability gap has a specific statistical cause when institutions target support at already-identified at-risk students, which is standard and sensible practice. Smith and Chimedza show by simulation that when treatment assignment depends on the variables that predict the outcome, a naive treated-versus-untreated comparison is biased, and that the size and direction of the bias can be verified only where the true effect is known by construction (Smith and Chimedza, 2021). This paper adopts that logic: Section 8 specifies an assignment mechanism deliberately confounded by prior-term observed risk, and Section 9 fixes a true intervention effect as a design constant, so any downstream causal analysis has a known correct answer to be checked against.

2.4 The Philippine Empirical Landscape

Three government and quasi-government sources anchor the calibration targets, and this paper states which figures come from which report. The House of Representatives' Congressional Policy and Budget Research Department (CPBRD) reproduces CHED's regional attrition figures for two consecutive graduating cohorts, with enrollment and institution counts per region, in its FY2026 Agency Budget Notes for CHED (CPBRD, 2025). The Second Congressional Commission on Education (EDCOM II) Year One report (2024) covers, among other findings, the targeting of tertiary subsidies and is not the source of the attrition figures used here. Its Year Two report (2025) is the source of the national and regional figures (≈39% nationally, and figures for BARMM, Region VII, Region IX, and CAR individually) that CPBRD's Agency Budget Notes later reproduce in its own narrative account of a legislator's floor speech, not as a CPBRD-computed statistic (EDCOM II, 2024, 2025). This provenance chain is central to the resolution of the national-attrition discrepancy in Section 10, and is why the EDCOM II figure and the CPBRD-quoted figure are not treated as independent corroborating sources. The Philippine Institute for Development Studies (PIDS) independently confirms the pandemic-era rise in national attrition and places it within broader concerns about CHED's regulatory and developmental capacity (Aldaba et al., 2024). Institutional qualitative work, including a 2025 phenomenological study of dropped-out students at a tuition-free public HEI, identifies financial struggles, family obligations, low academic readiness, and weak institutional support as drivers of dropout (Cahilog et al., 2025), with similar socioeconomic correlates in state-university research (Naz, 2024). Together these sources establish the aggregate scale of the phenomenon and its qualitative drivers, but none provides, because none could, the individual-level record that would let a researcher observe the hidden and recorded states of the same student at once.

3 Design Principles

Five principles govern the modeling decisions in the following sections, so each choice can be checked against a stated intention.

- **Empirical Fidelity:** every aggregate quantity the generator produces that has a real published counterpart (a national attrition rate, a regional split, a driver's relative prevalence) must reproduce it within ±3 percentage points (Section 11), and the counterpart's source must be citable. Two quantities are handled differently (Sections 10 and 11): the pooled national rate is derived from the tier targets and not matched to a national headline figure, and the held-out tier ratio has its own pre-specified band.
- **Deliberate Opacity:** the gap between what is recorded and what is true is modeled directly, not left as an incompleteness, so a downstream analyst working only from the exported tables faces the same informational limits as a real Student Success Analyst (Section 1).
- **Full Traceability:** every parameter is tagged as fixed by design, calibrated against a cited source, or explicitly assumed; none may carry an unstated status.
- **Built-In Falsifiability:** the model fixes at least one quantity, the true effect of the support intervention, that is unknowable in any real dataset but known here by construction, so a downstream causal analysis can be scored against a correct answer rather than only a plausible-looking one.
- **Bounded Scope:** granularity is capped at what the cited statistics support. Where a real number exists only at the national or two-tier regional level, the model does not manufacture false precision by fitting seventeen individual regions to it.

4 System Architecture and End-to-End Pipeline

Six of the model's eight tables are panels with one row per student per academic term; Regions and Students are fixed at generation time. The choice is substantive: "will this student leave next term" concerns a sequence of states, so the data must contain the sequence as a sequence. This is the person-period (discrete-time panel) representation standard for discrete-time survival analysis, in which a term is the unit of time and a dropout is an observed event or, for students still enrolled at the end of the observation window, a right-censored non-event.

Generation proceeds through five phases (Fig. 1). Phases 1–4 and the per-term draw in Phase 5 (Phase 5a) run in fixed order over each student's full enrollment history. Phase 5's aggregate check against real benchmarks (Phase 5b) is a batch-level operation that can run only once all N students exist; its output recalibrates the numeric parameters set in Phases 1 and 2 before the generator is final (the dashed feedback path in Fig. 1). This closed loop of parameter setting, output generation, and comparison with real benchmarks follows standard input-modeling and verification-and-validation practice for discrete-event and Monte Carlo simulation, in which a model is not accepted until its output has been checked against real-world reference data (Law, 2015). Because the loop tunes parameters against the benchmarks it is then checked against, it is calibration-until-convergence, not independent held-out validation (Section 11).

Table 2 lists the eight output tables. Field-level schemas and data types are in Appendix B. Appendix A gives one admissible instantiation of the generative forms for the observed variables, used to check the specification in a pilot; both appendices are kept out of the main text so implementation detail does not obscure the architecture. An implementation may differ from Appendix A provided it respects the constraints of Sections 5–9.

The generator is specified to produce 50,000 students [LOCKED; a scale decision, not a calibrated one], each followed for up to eight terms. This size keeps the tier-level validation check of Section 11 statistically meaningful although the high-attrition tier holds only about 15% of the population (Section 10): the tier still contains roughly 7,500 students, and a binomial back-of-envelope calculation puts the sampling error of a simulated attrition rate near 67% at about half

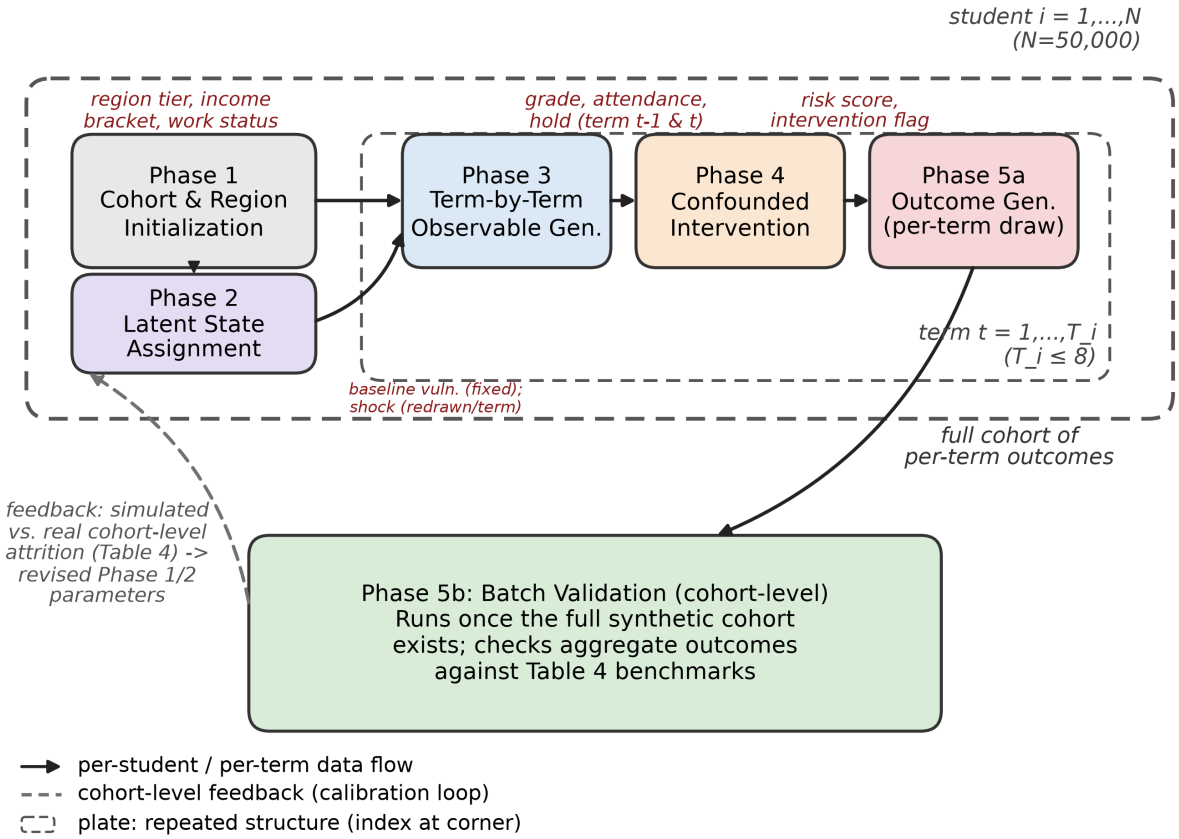


Figure 1: End-to-end synthetic cohort generation pipeline. The outer plate (student $i = 1, \ldots, N$, $N = 50{,}000$) and inner plate (term $t = 1, \ldots, T_i$, $T_i \leq 8$) show the two levels of repetition. Phases 1–4 and the per-term outcome draw (5a) repeat inside these plates; batch validation (5b) is drawn outside both because it is a cohort-level operation that runs once, after the full cohort exists. The dashed arrow is the calibration feedback path, not a data-generation step.

a percentage point ($\sqrt{0.67 \times 0.33/7{,}500} \approx 0.0054$). This is an order-of-magnitude check, not a precise bound: CPBRD enrollment data place the 15% tier share within a citable range (Table 4) but pin it down only to 14.1–18%, so a two-decimal error figure would overstate precision. The robust conclusion is that sampling noise at $N = 50{,}000$ is an order of magnitude smaller than the ± 3-percentage-point tolerance of Section 11 across that whole range, so a validation failure at that threshold would reflect genuine miscalibration, not sampling noise.

5 Phase 1: Cohort and Region Initialization

Every synthetic student begins with fixed attributes drawn independently of any later state: region tier, entry year, program discipline, household income bracket, working-student status, distance from campus, and subsidy status at entry. Region tier is the most consequential, because it connects the model to the largest and most reliable real benchmark available: published regional attrition figures.

Reported four-year cumulative attrition varies widely across regions within each cohort: CPBRD's regional table reports 36.3% to 93.4% for the Class of 2022 and 8.0% to 38.4% for the Class of 2023 (BARMM unreported), with a fall of about 19 percentage points on average per region from

Table 1: Generation pipeline phases

Phase	Activity
Phase 1: Cohort and Region Initialization	Each synthetic student is assigned a region tier, entry year, program discipline, income bracket, working-student status, and other fixed entry attributes. These are the only quantities generated before any latent state exists.
Phase 2: Latent State Assignment	A fixed baseline vulnerability is drawn as a function of the Phase 1 attributes. A per-term shock indicator is drawn independently in every subsequent term. Neither is exported.
Phase 3: Term-by-Term Observable Generation	For each term in sequence, term grade, attendance rate, and financial-hold status are drawn as noisy functions of the current baseline vulnerability and that term's shock.
Phase 4: Confounded Intervention Assignment	From a student's second term onward, an observed risk score is computed from the prior term's grade, attendance, and financial-hold values only. Support is received this term with that score as its probability (Eq. 1–2); no intervention is assigned in term 1, which has no prior term (Section 8).
Phase 5a: Outcome Generation (per student-term)	Dropout this term is a Bernoulli trial (Eq. 3–4) whose probability is the hazard of Eq. 3, a function of baseline vulnerability, the current shock, and whether the student received support this term.
Phase 5b: Batch Validation (per cohort, once)	After the full cohort of N students exists, aggregate outcomes are checked against the real benchmarks in Table 5 before the parameters are finalized (Section 11).

the first cohort to the second (CPBRD, 2025). Modeling every region would require region-specific calibration data this project lacks and would imply precision the sources cannot support, so the model groups regions into two tiers. The high-attrition tier comprises the four regions with the highest Class-of-2022 cumulative attrition in CPBRD's table (CPBRD, 2025): BARMM at 93.4%, Region VII at 60.7%, Region IX at 59.5%, and CAR at 54.9%, all well above the 36.3% (Region II) at the low end of that column. (Region IX's 59.5% is the figure in CPBRD's infographic table; the same document's narrative text states 59.9% when reporting a legislator's floor speech, a source-internal inconsistency this paper did not create and resolves by using the tabulated figure throughout; see Section 10.) EDCOM II's Year Two report independently reproduces the four tabulated figures exactly, and does not contain the 59.9% figure at all (EDCOM II, 2025). This strengthens confidence in the tabulated regional figures, though, as Section 10 discusses, not in the national total. The typical tier comprises all remaining regions (thirteen, all with reported values). The grouping is a deliberately coarse stand-in for region-level random effects nested within a national distribution (Gelman et al., 2013), a scope decision revisited in Section 14. Section 10 gives each tier's numeric target.

Table 2: Entity inventory

Table	Grain	Contents
Regions	one per tier	Reference values used to calibrate regional attrition targets (Section 10).
Students	one per student	Fixed entry attributes: region, entry year, program, income bracket, working-student status, distance from campus, subsidy status, and the latent baseline vulnerability (not exported).
Enrollments	student × term	Enrollment status and the end-of-window censoring flag.
CourseGrades	student × term	Term average score (0–100 scale), units attempted and passed.
AttendanceRecords	student × term	Attendance rate for the term.
FinancialHolds	student × term	Financial-hold status, overdue amount, and subsidy status for the term.
SupportInterventions	student × term	The observed risk score (undefined in a student's first term) and whether an intervention was received (a single undifferentiated flag, always 0 in term 1; Section 14 states why no intervention-type field is included).
PersistenceOutcomes	student × term	The dropout flag for the term and the rolled-up cohort attrition flag used for validation.

6 Phase 2: Latent State Assignment

Two quantities generated in this phase are never exported to any table and are inaccessible to any analysis of the dataset. They are the true, unobserved cause of everything that follows.

Baseline vulnerability is a continuous value fixed once per student at entry as a function of region tier, income bracket, and working-student status. It represents structural exposure to the pressures Section 2 identifies as dominant drivers of Philippine attrition, namely financial precarity and competing employment demands, and is unobservable just as those pressures are not directly recorded in any real student-information system.

Term shock is a per-term indicator, redrawn independently every term with probability π_s (Table 4), representing an acute disruption uncorrelated with the student's baseline standing: a medical event, a family emergency, an unplanned expense. Its independence from baseline vulnerability is a load-bearing design choice: it lets a student with low baseline vulnerability and no adverse recorded history still leave, which is the mechanism behind the first edge case in Section 12. Fig. 2 shows the causal structure connecting these two latent quantities to every observed and derived quantity in Sections 7–9, drawn in the probabilistic-graphical-model convention of the Bayesian modeling literature, which distinguishes random variables from deterministic functions and observed from latent quantities (McElreath, 2020; Gelman et al., 2013).

7 Phase 3: Term-by-Term Observable Generation

Term grade, attendance rate, and financial-hold status are each a noisy function of the current baseline vulnerability and the current term's shock. The noise is functionally necessary: it keeps

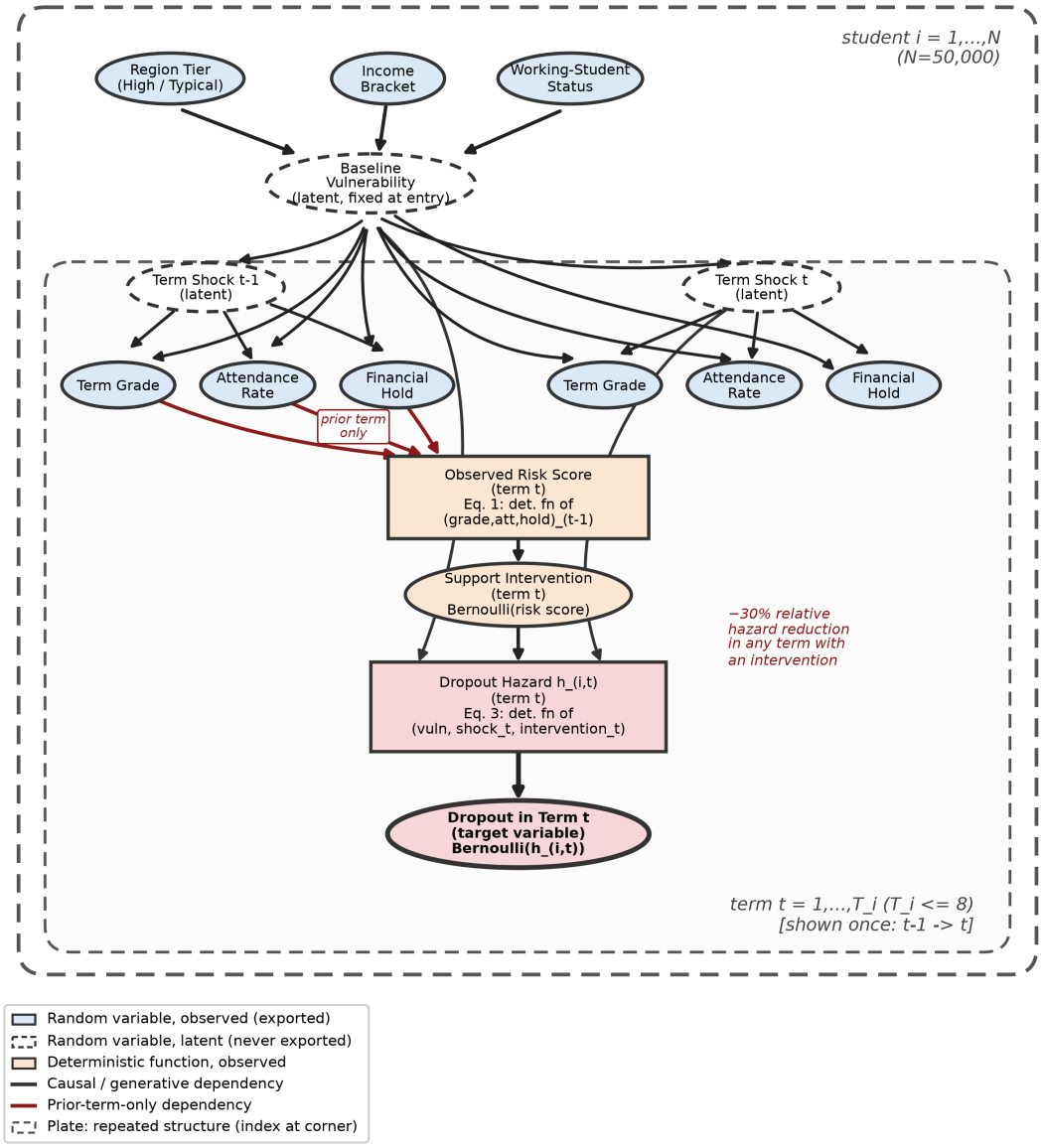


Figure 2: Latent and observed causal structure of the generative model. Following standard PGM convention, circles/ellipses denote random variables and rectangles denote deterministic functions of their parents; shaded (colored) nodes are exported to the dataset, and white nodes with dashed outlines are latent and never exported. The outer plate is the per-student repetition ($i = 1, \ldots, N$); the inner plate is the per-term repetition ($t = 1, \ldots, T_i$, $T_i \leq 8$), shown unrolled once across an adjacent term pair $t-1 \to t$ to make the prior-term-only dependency (red edges) visible. The whole shaded band recurs every term: term t's own Term Grade, Attendance Rate, and Financial Hold nodes are the "term $t-1$" inputs to Eq. 1 in the next repetition, closing a loop of which the unrolled pair shows only one link. Node fill colors match the phase color-coding of Fig. 1: purple for Phase 2 latent state, blue for Phase 3 observables, orange for the Phase 4 risk score and intervention, and red/pink for the Phase 5 hazard and dropout outcome. The Dropout Hazard node makes Eq. 3 visually explicit.

the observed variables from fully revealing the latent state, which is what makes the downstream analyst's problem nontrivial rather than a disguised lookup. Two students with identical baseline vulnerability and shock status will in general present differently on these variables, and two who present identically will in general differ in true vulnerability.

The three variables are independent of each other conditional on the latent state but correlated unconditionally, because they share the same latent cause. This mirrors real administrative data, where attendance and financial-hold status are often mildly correlated not because one causes the other but because both follow from a common, unrecorded circumstance.

8 Phase 4: Confounded Intervention Assignment

Throughout this section and Section 9, $i = 1, \ldots, N$ indexes students ($N = 50{,}000$, Table 2) and $t = 1, \ldots, T_i$ indexes terms, where $T_i \leq 8$ is the term in which student i drops out or, for students still enrolled, the last term of the observation window (the right-censoring point of Section 4). $\sigma(x) = 1/(1 + e^{-x})$ is the standard logistic function, mapping any real input to $(0, 1)$ so it can be read as a probability.

An observed risk score is computed from the prior term's grade, attendance, and financial-hold values, never from the current term's values or the latent state. An institution deciding whether to intervene this term cannot see this term's outcomes or the latent state, and the mechanism faces the same constraint.

In term $t = 1$ there is no prior term, so no risk score exists. Eqs. 1–2 are therefore defined for $t \geq 2$ only, and $\text{intervention}_{i,1} = 0$ for every student, so no support decision rests on a record that does not yet exist. Entry attributes (Section 5) are deliberately not substituted as a term-1 input: that would add a second path into assignment outside the three prior-term signals and change what an adjusted analysis must condition on (Table 5). This is a simplification, since a real institution may act on entry information, which this version does not represent. Consequently a student's first term is always untreated (the term-1 hazard in Eq. 3 carries no intervention factor), at most seven of eight terms are eligible for support, and every statement below about targeting rates, the eight-valued score, and the Confounding Presence and Recoverability benchmarks refers to student-terms with $t \geq 2$.

One defensible form for the score is a logistic combination of prior-term indicators:

$$\text{risk}_{i,t} \;=\; \sigma\Big(\beta_0 + \beta_1 \mathbf{1}[\text{grade}_{i,t-1} < \tau_g] + \beta_2 \mathbf{1}[\text{attendance}_{i,t-1} < \tau_a] + \beta_3\, \text{hold}_{i,t-1}\Big), \quad \beta_1, \beta_2, \beta_3 > 0 \tag{1}$$

Here $\text{hold}_{i,t-1}$ is the prior term's binary financial-hold flag. The sign constraint $\beta_1, \beta_2, \beta_3 > 0$ is enforced in the generator, so the direction of the risk relationship (a low grade, low attendance, or an active hold each independently *raises* risk) is a mathematical property of Eq. 1, not a prose claim. The thresholds τ_g and τ_a mark what counts as low; they take illustrative values in Table 4 and, like the three weights and β_0, are ASSUMED. This is the only functional form the main text gives for a Phase 3-derived quantity. The forms for grade, attendance, and hold (Section 7) are left open by the architecture, with one admissible instantiation in Appendix A, so the main text argues the architecture without serving as an implementation manual.

Support this term is a Bernoulli trial with $\text{risk}_{i,t}$ as the success probability:

$$\text{intervention}_{i,t} \mid \text{risk}_{i,t} \;\sim\; \text{Bernoulli}(\text{risk}_{i,t}) \tag{2}$$

Assignment is stochastic rather than a hard threshold rule (flag when $\text{risk}_{i,t}$ exceeds a cutoff θ) because the score is discrete. All three inputs to Eq. 1 are binary, so $\text{risk}_{i,t}$ takes only $2^3 = 8$

values across the up to 350,000 student-terms for which it is defined (50,000 students $\times$ up to 7 terms from $t = 2$; the true count is somewhat smaller after right-censoring, since not every student reaches term 8, Section 4). A hard cutoff can only partition those eight buckets, so it cannot hit an exact targeting rate except by coincidence of the population proportions. A stochastic draw can hit a target rate *in expectation* by giving each bucket its own intervention probability, which Eq. 2 does implicitly through $\sigma(\cdot)$. No threshold θ is needed.

A question of identification remains. With $\beta_0, \beta_1, \beta_2, \beta_3, \tau_g, \tau_a$ as six free quantities and only two aggregate targets (a targeting sensitivity of approximately 70% and a false-flag rate of approximately 25%, defined below), the system is underdetermined, and no solution is privileged without an additional assumption. This paper adds one explicit, ASSUMED simplification: the three weights are equal, $\beta_1 = \beta_2 = \beta_3 = \beta$, because no prior-term signal has a principled claim to more trust than the others at this scope. With τ_g and τ_a fixed at their illustrative Table 4 values (set independently of the targeting calculation, since they define what "low" means rather than how strongly a low reading counts), two free parameters, β_0 and β, face two targets. Table 3 and the discussion after it show what this does and does not pin down.

To confirm this is more than a labeling exercise, Table 3 works an illustrative case, purely to check that Eqs. 1–2 *can* reproduce the targeting figures. Suppose 30% of student-terms are genuinely high-current-vulnerability (an illustrative split, not a Table 4 parameter, since the real joint distribution of grade, attendance, and hold depends on the Phase 3 instantiation) and each of the three binary signals independently fires with probability 0.65 in that group versus 0.20 otherwise. Let $k \in \{0, 1, 2, 3\}$ be the number of signals that fire for a student-term; under equal weights risk depends only on k. Choosing (β_0, β) so the flagged population comes close to both targets gives, for example, $\beta_0 \approx -9$, $\beta \approx 6$:

Table 3: Worked illustrative parameter choice for Eqs. 1–2 under the equal-weight assumption ($\beta_0 \approx -9$, $\beta \approx 6$)

signals firing (k)	risk(k) $= \sigma(\beta_0 + \beta k)$	$P(k \mid$ high-vuln.)	$P(k \mid$ low-vuln.)
0	< 0.01	0.04	0.51
1	0.05	0.24	0.38
2	0.95	0.44	0.10
3	> 0.99	0.28	0.01

Averaging risk(k) against these two distributions gives a targeting sensitivity of approximately 71% and a false-flag rate of approximately 28%, close to the 70%/25% design targets and within what the Phase 5b calibration loop (Section 11) would fine-tune. Hitting both exactly is possible, but only with a near-step assignment, risk(0) $\approx$ risk(1) ≈ 0, risk(2) ≈ 0.96, risk(3) ≈ 1, which any sufficiently large β reproduces with $\beta_0 \approx 3.15 - 2\beta$ (for example $\beta \approx 10$ or $\beta \approx 37$). In this submodel the two targets therefore fix the assignment probabilities at $k = 2$ and $k = 3$, not a unique (β_0, β); the steepness is a free degree of freedom that the smoother pair in Table 3 resolves by choice. The exercise also yields a finding, not only a confirmation: the calibrated risk(k) values are close to a step function (near zero for $k \leq 1$, near one for $k \geq 2$), a direct consequence of hitting precise aggregate targets from a score with only eight possible values, the discreteness problem noted in Section 14. A second consequence concerns identification. With risk(0) < 0.01 and risk(3) > 0.99, assignment is nearly deterministic when no signal or all three signals fire (treated terms in the $k = 0$ cell, and untreated terms in the $k = 3$ cell, occur at roughly 1 in 8,000, since $\sigma(-9) \approx 1.2 \times 10^{-4}$), so only the one- and two-signal cells contain both treated and untreated student-terms. This is a near-violation of the positivity (overlap) condition that any adjusted analysis must respect (Section 11;

Appendix A reports pilot cell counts). A formally stochastic mechanism does not remove it; it only makes the targets achievable in expectation rather than by coincidence, while most variation across the eight buckets stays at the extremes.

The targeting is imperfect by design: approximately seventy percent of students with genuinely high current vulnerability are correctly identified and offered support, while approximately twenty-five percent of students offered support have genuinely low underlying vulnerability and were flagged because of a single weak term. This imperfection creates the confounding of Section 2.3 (Smith and Chimedza, 2021): assignment is correlated with, but not identical to, true risk, so a naive comparison of supported and unsupported students is an unreliable estimate of the program's effect. In observational-study terms, this violates the ignorable-treatment-assignment assumption a simple group comparison needs to identify a causal effect (Gelman et al., 2020), and a corrected analysis must condition away exactly this confounding path in the causal-inference sense (Peters et al., 2017). This is the failure mode the dataset's intervention-effectiveness component lets an analyst discover and correct for.

9 Phase 5: Outcome Generation and Validation

Section 4 frames dropout as a discrete-time hazard problem. This section states the hazard equation, the paper's most important functional form, because it generates the target variable. Dropout in a given term is a Bernoulli trial whose success probability is a discrete-time hazard, a deterministic logistic function of baseline vulnerability, that term's shock, and whether the student received support this term:

$$h_{i,t} = \sigma(\gamma_0 + \gamma_1\, v_i + \gamma_2\, \text{shock}_{i,t}) \times (1 - 0.30 \cdot \text{intervention}_{i,t})\,, \qquad \gamma_1, \gamma_2 > 0 \tag{3}$$

$$\text{dropout}_{i,t} \mid h_{i,t} \sim \text{Bernoulli}(h_{i,t}) \tag{4}$$

for students still enrolled at the start of term t (i.e., $\text{dropout}_{i,t'} = 0$ for all $t' < t$), where v_i is baseline vulnerability standardized to a continuous scale and $\text{shock}_{i,t} \in \{0, 1\}$ is that term's shock indicator (Section 6).

Unlike β_0, β in Section 8, $\gamma_0, \gamma_1, \gamma_2$ are not given point values here. Eq. 3's three parameters face the same underdetermination that Section 8 resolved for Eq. 1's six, and the architecture offers no analogous assumption; Appendix A resolves it by explicit choice. The two independent calibration targets are the tier-level cumulative attrition rates in Table 4, both Class of 2022: high tier ≈67%, typical tier ≈46%. (The pooled national figure, ≈49% at the 15% tier share, is their share-weighted blend, not a third constraint.) Two numbers cannot identify three parameters without a further assumption. Section 8's equal-weighting assumption has no analogue here: γ_1, a coefficient on continuous standardized vulnerability, and γ_2, a coefficient on a binary shock, are not structurally interchangeable, so fixing their ratio would be unmotivated. A third kind of target would resolve the remaining degree of freedom: a term-by-term hazard shape (whether risk within a tier is concentrated early, late, or roughly uniform). The cumulative figures of Section 10 describe only the four-year total, and no source cited in Section 2 reports Philippine attrition at term level, so that target is unavailable. Fitting $(\gamma_0, \gamma_1, \gamma_2)$ is therefore a multi-term simulation-calibration exercise run by the Phase 1/2 feedback loop (Section 4) against the two tier targets. Appendix A resolves the remaining degree of freedom by fixing γ_1 and γ_2 as ASSUMED values and calibrating γ_0, together with the tier shift in baseline vulnerability, to the two targets; this is one admissible resolution, not one the architecture requires.

The 0.30 factor in Eq. 3 applies to every student-term with $\text{intervention}_{i,t} = 1$. It is an internal design constant (LOCKED, Table 4), not a claim about any real intervention's effectiveness, so

that a downstream causal analysis has a specific, known value to recover rather than a plausible range to argue for. The equation has no compliance or fidelity gate: the model does not distinguish an administered intervention that "genuinely" helped from one that did not, and every such student-term receives the same 0.30 reduction by construction. The 30% reduction is thus a fixed, population-average effect, not a per-student heterogeneous effect drawn from a further latent distribution. This keeps the Recoverability benchmark in Table 5 well posed: "recovers an estimate close to the true effect" has one unambiguous target. A heterogeneous-effects version, in which a correctly specified analysis must recover a conditional average treatment effect rather than a single number, is a recognized and active extension of this simulation-based validation approach (Smith et al., 2020), left as future work (Section 14).

On the magnitude of 30% itself: no real Philippine study of institutional retention-support interventions was located that would let it be calibrated like the attrition figures in Table 4, so it remains ASSUMED-in-spirit despite the LOCKED tag (LOCKED means fixed by design for internal consistency, not empirically settled). As a rough plausibility check, not a calibration source, a 2024 meta-analysis of dropout-intervention programs finds effect sizes of about 0.31 for combined and 0.33 for behavioral interventions (Wang, 2024). That study is K–12 and reports an effect size rather than a hazard-ratio reduction, so the comparison is order-of-magnitude only; the effects are modest rather than transformative, the same register as a 30% relative hazard reduction, which is large enough to be a checkable signal for the Recoverability benchmark but not so large that an analyst could recover it by accident from noise. A closer match on the metric, though not the population, is a randomized trial of a middle-school relevance intervention whose event-history analysis found a 23% reduction in the hazard of later high-school dropout (Rose and Bowen, 2021); it too is K–12 and a single study. This paper does not claim these figures are comparable to the effect of a higher-education support program, only that 30% is not an implausible order of magnitude for this literature. The value remains a design choice made for the validation goal of Section 11, not an empirical estimate.

At the end of each student's observed history, term-level dropout flags are rolled up into one cohort-level attrition flag comparable in definition to the published four-year cumulative figures (Section 5). This rolled-up quantity, not the per-term hazard, is checked against the real benchmarks in Table 5: per-term mechanics are internal, but aggregate behavior must match cited reality within ±3 percentage points (Section 11) before the parameter set is final. Otherwise Phase 1 and Phase 2 parameters are revised and the cohort regenerated (the feedback path in Fig. 1).

10 Calibration Parameter Design

Every numeric parameter carries exactly one of three tags. LOCKED denotes a value fixed by design and not intended to be tuned. CALIBRATED denotes a value set to reproduce a specific, cited figure. ASSUMED denotes a defensible modeling choice made without a directly applicable statistic, flagged as a candidate for future sensitivity analysis rather than presented as settled. Table 4 lists the parameters with the greatest influence on aggregate behavior; a complete parameter log accompanies the generator implementation.

Table 4: Selected calibration parameters

Parameter	Value	Tag	Basis
National 4-year cumulative attrition, Class of 2023 (held-out level target only)	≈ 29.4%	CALIBRATED	CHED's national total in CPBRD's regional table, printed without a parenthetical (CPBRD, 2025); read as the Class of 2023 figure (see below) and used only to fix the overall level in the held-out check (Section 11), never as a Class of 2022 target.
High-attrition tier 4-year cumulative target (Class of 2022)	≈ 67%	CALIBRATED	Simple average of the four regions with the highest Class-of-2022 cumulative attrition in (CPBRD, 2025): BARMM, Region VII, Region IX, CAR (simple rather than enrollment-weighted, because BARMM's enrollment is not reported; the effect is quantified below); independently reproduced exactly (including Region IX at 59.5%) in (EDCOM II, 2025). See Section 5 on the separate 59.9% figure unique to CPBRD's narrative text.
Typical tier 4-year cumulative target (Class of 2022)	≈ 46%	CALIBRATED	Enrollment-weighted mean of the Class-of-2022 values for the thirteen regions outside the high tier (CPBRD, 2025), using each region's SY 2023–24 enrollment from the same table as a proxy for cohort size (45.9%; the unweighted mean is 44.7% and the thirteen values span 36.3%–52.4%, so the ±3-point tolerance in Section 11 covers either choice).
Share of population in high-attrition tier	15%	ASSUMED	Not set to reproduce a cited figure, but within a citable range: CPBRD's SY'23–'24 enrollment (CPBRD, 2025) gives CAR + Region VII + Region IX at 149,843 + 407,022 + 168,298 = 725,163 of a national total of 5,140,409 students (14.1%). BARMM's enrollment is unreported, and that total is the exact sum of the sixteen regions with reported enrollment, so it excludes BARMM; adding BARMM to numerator and denominator can only raise the share, plausibly to 14.1–18%. The 15% was chosen before this calculation and sits inside that range.
Working-student prevalence	35%	ASSUMED	Directionally supported by survey evidence that employment is a leading deterrent to continued enrollment; not itself a matched statistic.

continued on next page

Table 4 continued

Parameter	Value	Tag	Basis
Term shock probability (per student-term, π_s)	8%	ASSUMED	Illustrative per-term rate of an acute disruption (medical event, family emergency, unplanned expense); not calibrated to a specific source. Structurally a simple population-level Bernoulli rate, like working-student prevalence above, and listed for the same reason.
Grade threshold in Eq. 1 (τ_g)	60 (of 100)	ASSUMED	Illustrative cutoff marking a low term grade on the model's 0–100 term-average-score scale; not calibrated to a specific source.
Attendance threshold in Eq. 1 (τ_a)	75%	ASSUMED	Illustrative cutoff marking low attendance; not calibrated to a specific source.
Logistic intercept β_0 in Eq. 1	≈ -9	ASSUMED	Chosen jointly with β below, under an illustrative population-separation submodel, to approximate the targeting figures below (the targets alone do not fix a unique pair); see the worked example in Section 8 (Table 3). Used unchanged in the Appendix A pilot rather than fitted to any simulated population.
Shared prior-term signal weight $\beta = \beta_1 = \beta_2 = \beta_3$ in Eq. 1	≈ 6	ASSUMED	Equal weighting across grade/attendance/hold imposed as a symmetry simplification (Section 8); chosen jointly with β_0 above.
Intervention targeting precision (true positives)	70%	ASSUMED	A deliberately imperfect value; see Section 8.
Intervention false-flag rate	25%	ASSUMED	A deliberately imperfect value; see Section 8 and the second edge case in Section 12.
True intervention effect (relative hazard reduction)	30%	LOCKED	An internal ground-truth constant by design, not an empirical claim; order-of-magnitude consistent with real dropout-intervention meta-analyses (Wang, 2024); see Section 9.
Term granularity	1 term = 1 semester; 8 terms per 4-year cohort	LOCKED	Fixed by design to match the standard Philippine academic calendar.

CPBRD's FY2026 Agency Budget Notes for CHED contain two national attrition figures. They can look like competing estimates of one quantity. Tracing each to its origin and graduating cohort shows they are not, and that pairing one cohort's national figure with another cohort's regional figures would mix cohorts. The discrepancy is resolved here, with its reasoning shown.

The ≈39% figure. This is the figure most often repeated in press coverage, and it appears in

both CPBRD's document and an EDCOM II report, which might make it look well corroborated. Tracing it removes that appearance. Within the CPBRD document, the whole cluster built around 39% (national ≈39%, BARMM 93.4%, Region VII 60.7%, Region IX 59.9%, CAR 54.9%) is not a CPBRD-computed statistic. It appears in a narrative paragraph that CPBRD attributes to Representative Jude Acidre's privilege speech of September 1, 2025 (CPBRD, 2025) (the Highlights section repeats the same sentences without attribution). The passage has no quotation marks and reads as CPBRD's account of what the speech said, so this paper describes it as CPBRD reporting the speech's content, not quoting it; the speech transcript was not independently located to confirm word-for-word agreement. The same cluster, with Region IX at 59.5%, already appears in EDCOM II's Year Two report of January 2025, eight months before the speech (EDCOM II, 2025), and Representative Acidre is an EDCOM II commissioner. The most parsimonious reading is that the speech restates EDCOM II's own figure, so a legislator citing a report he co-authored is not a second corroborating source. The 59.9% Region IX figure in the narrative cluster also differs from the 59.5% in CPBRD's own regional table for the same region and cohort (Section 5), a second reason to treat the narrative cluster with more caution than the tabulated data.

The 29.4% figure. This appears in CPBRD's infographic table as the national total, attributed explicitly and only to "CHED Office of Planning, Research, and Knowledge Management" (CPBRD, 2025), CHED's primary internal data source, with a short, direct, named chain of custody. The table's attrition column is headed "Class of '23 ('22)": each region carries the Class of 2023 value first and the Class of 2022 value in parentheses, and the national row carries a single unparenthesized value. This paper reads it as the Class of 2023 national figure. The reading is inferred from the table's layout, not printed, and an independent check supports it: weighting each region's value by SY 2023–24 enrollment from the same table (a proxy for cohort size; BARMM excluded because neither its enrollment nor its Class of 2023 value is reported), the Class of 2023 column averages 27.4%, within about two points of 29.4%, whereas the Class of 2022 column averages 47.8%. The same table shows that the Class of 2022 regional values quoted alongside 39% do not reproduce it: the sixteen reported values average 47.8% enrollment-weighted and 47.3% unweighted, about nine points above 39%, and neither CPBRD nor EDCOM II states the batch or base population of the 39%. Two further sources bear on cohort. CHED's FY2026 budget presentation to the House reports 29.39% bachelor's-degree attrition for AY 2023–24 (Commission on Higher Education, 2025), matching the CPBRD national total and supporting its reading as the more recent cohort. A press report of CHED data given to the House in 2023 gives national attrition of 38.95% for the SY 2021–22 graduating batch and 41.16% for the 2023 batch, with CHED defining attrition as students who temporarily leave or drop out permanently (Chi, 2023). The first is a candidate underlying source for the 39% figure, but neither CPBRD nor EDCOM II cites it, and CPBRD's "Class of '22" label is not tied to a school year, so these documents cannot establish whether the 39% and the Class of 2022 regional values describe the same batch. The chain of custody for 39% in CPBRD and EDCOM II therefore remains indirect: a floor speech, then a commission report that does not cite its source for the number.

Resolution. The model uses neither national figure as a Class of 2022 target. It calibrates by tier, with both targets from the same cohort and column of the same table (Table 4): ≈67% for the high-attrition tier and ≈46% for the typical tier. The pooled national rate is derived, ≈49% at the assumed 15% tier share, and reported but not matched to an external figure. The only national figure found quoted alongside the Class of 2022 regional values is the 39% figure, so the derived rate sits about ten percentage points above it. This gap is a known, unresolved consequence of calibrating to the regional table rather than a national headline: reproducing 39% with a 15% high tier at ≈67% would require a typical-tier rate of about 34% (33–34% across the 14.1–18% share range), below every reported Class of 2022 regional value, the lowest being 36.3%. The 29.4%

figure is retained, as a Class of 2023 figure, only to fix the overall level in the held-out check of Section 11. The 39% figure is logged for transparency, is not a calibration target, and is not treated as corroborated by repetition across CPBRD and EDCOM II, which are not independent. The choice matters: blending a Class of 2022 high tier at ≈67% with the Class of 2023 national figure would imply a typical-tier target of ≈23%, which no Class of 2022 regional value supports (the lowest is 36.3%), and a high-to-typical ratio of about 2.9 where the Class of 2022 column supports about 1.5 (67.1% to 45.9%). A future revision should revisit this if CHED source data with a documented cohort definition and base population for each national figure become available.

BARMM sensitivity. The high-tier target of 67.1% is a simple mean in which BARMM, whose enrollment the table does not report, carries a quarter of the weight. The three high-tier regions with reported enrollment average 59.2% over 725,163 students; adding BARMM at 93.4% to an enrollment-weighted mean gives 60.4% if BARMM enrolls 25,000 students, 63.4% at 100,000, 65.1% at 150,000, and 68.0% at 250,000. For any BARMM enrollment below about 121,000 the weighted target would fall below the lower edge of the ±3-point band around 67.1% (64.1%), and the high-to-typical ratio would be nearer 1.3–1.4 than 1.5. The simple mean is kept because it uses only the four cited regional values and needs no BARMM enrollment figure. It coincides with the enrollment-weighted mean only if BARMM enrolls about 218,000 students, which this paper has no data to confirm or rule out, so the enrollment-weighted alternative is a candidate for the sensitivity analysis that the tagging above calls for.

11 Validation Measures and Success Benchmarks

Completeness for the generator means its aggregate output satisfies every check in Table 5, not merely that it runs without error. The checks run after every parameter revision, not only at the end.

Estimand. In Eq. 3 the intervention multiplies the per-term dropout probability by 0.70, so the true effect is a risk ratio (equivalently, a ratio of discrete-time hazards) of exactly 0.70, and "30%" in Table 5 means $1 - RR$. Within any stratum of the prior-term signals, and under any common standardization across strata, the ratio is the same, because assignment is independent of the latent state given those signals and both arms share the same baseline hazard. It is not an odds ratio, and the treatment coefficient of a logistic regression does not estimate it: the odds ratio implied by a fixed risk ratio of 0.70 drifts away from 0.70 as the baseline hazard rises (about 0.67 at a per-term hazard of 0.13, roughly the average implied by 67% cumulative attrition over eight terms under a constant-hazard approximation, and about 0.64 at 0.25, an apparent 36% reduction already outside the 5-point tolerance). Recoverability is therefore scored on the risk ratio. An adjusted analysis should use an estimator that targets it directly and borrows strength across the prior-signal cells (for example, log-binomial or Poisson regression with robust standard errors, or a Mantel–Haenszel risk ratio across those cells). Inverse-probability weighting and full-population standardization are unsuitable: because assignment is nearly deterministic in two of the four signal-count cells (Section 8), their weights are extreme (up to about 8,000) and the cells they must extrapolate into contain almost no observations of one arm. A logistic model is acceptable only if its fitted probabilities are converted back to a risk ratio by model-based standardization. Because the effect is identified only from cells in which both arms are populated, a correct estimator's sampling variability at $N = 50{,}000$ can be comparable to the 5-point window in Table 5, so that window is set from replicated runs rather than assumed (the estimator's standard deviation is about 4.4 points in the pilot of Appendix A).

The national- and tier-level attrition checks are calibration-until-convergence, not independent validation. The feedback path of Fig. 1 (Section 4) tunes Phase 1/2 parameters against exactly

Table 5: Validation measures and success benchmarks

Measure	Benchmark
National cohort attrition (derived)	Simulated 4-year cumulative attrition, pooled across both tiers, falls within ±3 percentage points of the share-weighted blend of the two tier targets in Table 4 (≈49% at the 15% tier share, about ten points above the 39% headline figure). This is an internal-consistency check, not a comparison against an external national figure; see Section 10 for why neither the 29.4% nor the 39% figure is a Class of 2022 target.
Tier-level attrition	Each tier's simulated 4-year cumulative attrition falls within ±3 percentage points of its own Class of 2022 target in Table 4 (≈67% high, ≈46% typical).
Held-out cohort (Class of 2023)	With every calibrated parameter frozen except the overall hazard level, refit so that pooled simulated attrition is within 1 percentage point of 29.4% (the Class of 2023 national figure); the simulated ratio of high-tier to typical-tier cumulative attrition then lies between 1.2 and 1.7, bracketing the observed Class of 2023 ratio of 1.37. The procedure is specified in the text below.
Confounding presence	A naive (unadjusted) comparison of dropout rates between intervention recipients and non-recipients among student-terms with $t \geq 2$, reported as the estimated relative reduction $1 - \widehat{RR}$ with $\widehat{RR}$ the ratio of the two raw dropout rates, yields a point estimate whose 95% confidence interval excludes the true 30% relative-hazard-reduction effect (Eq. 3). This confirms the confounding was generated and is large enough to bias a naive estimate away from the true value rather than being accidentally cancelled out.
Recoverability	An analysis that conditions on the prior-term signals driving assignment (e.g., regression adjustment or a Mantel–Haenszel estimator on $\text{grade}_{i,t-1}, \text{attendance}_{i,t-1}, \text{hold}_{i,t-1}$, following standard practice for confounded observational treatment effects (Gelman et al., 2020)), restricted to student-terms with $t \geq 2$ and estimating the same quantity as Eq. 3 (the risk ratio; see the estimand note), yields a relative-reduction estimate $1 - \widehat{RR}$ whose 95% confidence interval includes the true 30% effect, and whose point estimate lies within the larger of 5 percentage points and twice the estimator's standard deviation across at least 20 independent runs of the generator (see the estimand note).
Edge-case presence	Both edge cases in Section 12 occur in the generated cohort at a non-trivial, auditable rate; they must arise from the generative mechanism, not be inserted as special-cased rows.
Parameter traceability	Every parameter in the generator implementation carries exactly one of the three tags in Section 10; none are left unmarked.

these benchmarks, so passing them shows the tuning converged, not that the model would reproduce a real statistic it was not tuned against, which is what held-out validation must show. A genuinely held-out check compares calibrated output against a reference figure never used in tuning. This specification pre-specifies one; Appendix A reports it for a pilot instantiation, not a production generator. The reference is the Class of 2023 column of the same CPBRD table (CPBRD, 2025),

which supplies no tier target; the Class of 2022 column is the calibration cohort. The check has three steps. First, calibrate to the Class of 2022 tier targets as above. Second, freeze every calibrated parameter except the overall hazard level (the intercept γ_0 of Eq. 3, which belongs to the Phase 5 hazard but is fitted by the same feedback loop) and refit it until pooled simulated attrition is within 1 percentage point of 29.4%, the Class of 2023 national figure. Third, without further tuning, compute the simulated ratio of high-tier to typical-tier cumulative attrition and require it to lie between 1.2 and 1.7.

The reference values are as follows. In the Class of 2023 column the three high-tier regions with a reported value (CAR, Region VII, Region IX; BARMM is unreported) average 35.7% and the thirteen typical regions 26.0%, both enrollment-weighted, a ratio of 1.37. On the same three regions the Class of 2022 ratio is 1.29, so the ratio moved by 0.08 between cohorts; including BARMM (with the simple four-region mean used as the tier target) raises the Class of 2022 ratio from 1.29 to 1.46, indicating how much the missing BARMM value could matter. The band runs from 1.37 less 0.15 (about twice the observed drift) to 1.37 plus 0.17 (the BARMM effect) plus 0.15, rounded to 1.2 and 1.7; the band edges and the 1-point refit tolerance are ASSUMED in the sense of Section 10. The check targets this ratio, not the level, because levels are unstable across cohorts: region-level attrition fell by about 19 percentage points on average between the two cohorts (standard deviation 8.6 points across the sixteen regions with both values, rank correlation only 0.39), so a level comparison at the ±3-point tolerance used elsewhere would measure cohort-to-cohort volatility, not model fidelity.

Two limits are stated rather than hidden. The check has little power: one structural quantity is predicted after fitting one level parameter, and the band is wide, so passing is weak evidence of generalization while failing would be informative. And it is not blind: the Class of 2023 values were inspected, to compute the reference ratio and the drift above, while the check was being designed, so it is held out from parameter fitting but not from design choices. The CPBRD table reports only two cohorts, so no further held-out cohort is available from this source. In the pilot instantiation the check lands marginally outside the band (Appendix A). The Confounding Presence and Recoverability checks are not subject to this calibration-loop limitation: the true 30% effect is fixed by design (Eq. 3), not fit from the data being checked, so comparing a naive and an adjusted estimator's recovery of it is an independent check in the ordinary sense.

The exported target variable (one row per student per term, with a binary dropout flag and an end-of-window censoring indicator) supports two complementary downstream analyses in a companion deliverable: supervised classification of the per-term dropout flag, of the kind catalogued in the statistical-learning literature (James et al., 2013), and a discrete-time hazard formulation built on the person-period structure of Section 4 and Eq. 3. Neither analysis is performed in this specification; the data model keeps both available.

12 Deliberate Edge Cases

Two edge cases are required outputs of the generative mechanism, not optional extras.

1. ***Unflagged high-risk student.*** A student with strong prior-term grades, good prior-term attendance, and no financial hold on record, and therefore a low observed risk score and no intervention this term, who nonetheless drops out because a term shock occurred that no institution could have acted on in time. This term's record is not itself blind to the shock: per Section 7, the shock also shapes this term's grade, attendance, and hold values, so in hindsight the data is not fully unrevealing. What is blind to it is the intervention decision, because Eq. 1 conditions only on the *prior* term's values (Section 8) and is computed before this term's shock and its consequences exist. This case follows directly from the independence

between term shock and the prior term's observed variables (Section 6). It shows that a support decision timed off recorded history cannot act on a disruption that has not happened yet, not that the disruption leaves no trace.

2. ***Falsely flagged student.*** A student referred for support on the basis of one weak term despite low genuine underlying vulnerability, consistent with the twenty-five percent false-flag rate of Section 8. This case shows that even a reasonably well-targeted early-warning mechanism allocates some real support resources to students who did not strictly need them, an operational cost worth surfacing in any downstream analysis rather than assuming clean targeting.

Both cases are checked for presence, not merely possibility, under the edge-case measure in Table 5.

13 Ethical and Responsible-Use Considerations

13.1 No Real Personal Data

Every student, term, grade, and event is synthetic. No real student's record is used as an input, template, or individual-level validation target; only aggregate, already-published regional and national statistics are used for calibration. Data-privacy obligations that attach to real student records (consent, retention limits, access control) do not apply to the output in the same way, because no real individual's information is represented.

13.2 Misrepresentation Risk

The principal ethical hazard here is misattribution, not privacy: presenting a pattern produced by this synthetic model as an empirical finding about real Philippine students. Any documentation, notebook, or presentation built on the model must state plainly, wherever a number derived from the synthetic data appears, that it describes the behavior of a generative model calibrated against cited aggregate statistics, not a measurement of any real population. The exception is the small set of input statistics in Table 4 marked CALIBRATED with a citation; those numbers are real and labeled as such.

13.3 Regional Framing

Region names appear in this document only attached to the real, cited figures published about them (Section 5, Table 4). No claim beyond those figures, about the cause of a region's attrition rate, the quality of its institutions, or the character of its students, is made or implied anywhere in the model, and any dissemination of work built on it should preserve that boundary.

This boundary is necessary but not sufficient. Calibrating a region's high-attrition-tier assignment to a cited figure is a claim about an aggregate statistic; it does not neutralize a separate risk one step downstream, once an implementation generates individual synthetic records. A generator calibrated on BARMM's 93.4% figure will by construction produce tens of thousands of individually plausible synthetic student narratives (a specific combination of low grade, poor attendance, an active financial hold, and eventual dropout) tagged to that real, specific, historically marginalized, post-conflict region. Aggregate calibration accuracy says nothing about whether it is responsible to generate, retain, or circulate that volume of synthetic "at-risk BARMM student" narratives, particularly disaggregated to region level where a reader could mistake a synthetic profile for a description of real students there. This risk is distinct from, and not resolved by, the misattribution risk of Section 13.2: a chart correctly labeled as synthetic can still read as stigmatizing at

the individual-record level if presented, browsed, or distributed as a set of real case studies. Any implementation of this model should therefore: (i) present synthetic individual-level records, if at all, only with an unambiguous, unavoidable synthetic-data label at the point of display, not only in an appendix or methods section; (ii) avoid disseminating region-disaggregated individual-level synthetic records beyond the aggregate statistics needed for the validation checks in Table 5, since the aggregates carry the calibration value and the individual records add nothing to it; (iii) avoid pairing a record's region tag with narrative embellishment (a name, an invented life story) beyond the exported tabular fields in Table 2, since embellishment turns a calibration artifact into something that reads as a case study of a real person; and (iv) if a presentation needs an illustrative individual example at all, prefer one from the typical tier, or explicitly justify why a region-tagged high-attrition-tier example is necessary. These are recommendations for any implementation, not requirements this document can enforce.

13.4 Scope of Applicability

This model is a methodological exercise and training artifact, distinct in kind from a real institutional early-warning system: it has not been validated against one and cannot substitute for one. Any operational system built with real student data would need its own ethical review, consent framework, and data-protection compliance, none of which this synthetic specification satisfies.

14 Limitations and Scope

The model's scope is bounded deliberately, and the boundaries are stated here rather than left implicit.

- **Two-tier regional structure:** seventeen regions are collapsed into two tiers; a future iteration with region-specific calibration data could model finer structure.
- **Single shock type:** term shock is one undifferentiated latent event; real acute disruptions (medical, financial, family) likely differ in frequency and severity, which this version does not distinguish.
- **Single combined intervention:** support is one undifferentiated flag rather than separate tutoring, financial counseling, and advising outreach, which likely have different true effect sizes.
- **Fixed, not heterogeneous, intervention effect:** Eq. 3 applies the same 30% relative hazard reduction to every supported student-term, although the paper's own cited related work on individual treatment effects (Smith et al., 2020) treats heterogeneity as the object of interest. This is a deliberate scope decision (Section 9) that gives the Recoverability benchmark (Table 5) one unambiguous value; a heterogeneous-effects version is a natural extension.
- **Discrete pre-noise risk score:** because Eq. 1's three inputs are binary, the pre-noise risk score takes only eight distinct values across the cohort ($2 \times 2 \times 2$ combinations of low grade, low attendance, and hold status), coarser than a real institutional score. Section 8's worked example shows this pushes calibrated assignment probabilities toward a near-step function even under a formally stochastic mechanism and makes assignment nearly deterministic in two of the four signal-count cells, limiting which estimators can recover the effect (Section 11). A later revision could add a continuous input, such as the overdue amount already tracked in FinancialHolds.
- **Cohort alignment of national figures:** Section 10 reads the 29.4% national figure as a

Class of 2023 value from the layout of CPBRD's table, supported by an enrollment-weighted check and by CHED's FY2026 budget presentation (which reports the same figure for AY 2023–24) but not by a printed label, and does not tie the 39% figure to a graduating batch. If the column order were read the other way, the calibration and held-out design of Sections 10 and 11 would need to be redone. The mapping of cohort labels to school years should be confirmed against CHED's underlying data. The typical-tier target of ≈46% also rests on enrollment weights that proxy, rather than equal, cohort sizes, and the derived national rate (≈49%) sits about ten points above the 39% headline figure.

- **Static, cohort-specific calibration:** regional attrition changes by about 19 percentage points on average between the Class of 2022 and Class of 2023. The model calibrates to one cohort (the Class of 2022, for which the regional tier structure is fully reported) rather than the most recent, and does not model trend, so its attrition levels are cohort-specific.
- **Attrition is not permanent dropout:** press coverage of CHED data describes CHED's attrition measure as counting students who temporarily leave as well as those who drop out permanently, computed from the change in enrollment between a batch's first and last years, and CPBRD's table defines it as drop-out rates for the whole cohort from first to fourth year. The model's dropout flag has no re-entry, so calibrating to these figures treats temporary stop-outs as dropouts; the calibrated rates are attrition, an upper bound on permanent exit.
- **Gender, disability, and indigenous status are not modeled:** the cited Philippine qualitative sources (Cahilog et al., 2025; Naz, 2024) describe financial and institutional drivers in detail but, in the studies cited, give no quantitative Philippine higher-education attrition breakdown by gender, disability, or indigenous identity that this model could calibrate against as Table 4 does for region and income; none was located in this project's source review. Their absence is an evidentiary gap in available calibration data, not a judgment that they are unimportant. A future iteration that locates or commissions such a breakdown should add them as Phase 1 attributes influencing baseline vulnerability (Section 6) rather than treating this version's silence as a finding.
- **Reproducibility stated, not guaranteed across platforms:** Appendix B states a seed and versioning policy for 50,000 students over up to eight terms each; bitwise reproducibility is expected only on a fixed platform and library version, and cross-platform equivalence is statistical, not exact.

15 Conclusion

This paper specifies a two-layer synthetic data model for student retention in Philippine higher education, in which a latent baseline vulnerability and an independent per-term shock jointly drive an exported record of grades, attendance, financial-hold status, and intervention history, without either latent quantity ever being visible to a downstream analyst. Aggregate behavior is calibrated against national and regional attrition figures from CPBRD, EDCOM II, and PIDS, traced to their sources and graduating cohorts in Section 10 rather than treated as one undifferentiated body of corroboration, and checked by the procedure of Section 11, which states where that procedure is calibration-until-convergence rather than independent validation and pre-specifies a limited held-out check against a second graduating cohort. The intervention-assignment mechanism is deliberately confounded and fully specified by two equations (Eqs. 1–2) with explicit index ranges, sign constraints, and a worked numeric demonstration, following the documented statistical failure mode of real early-warning systems, so the dataset carries a known ground-truth causal effect against which any downstream analysis can be scored rather than merely judged plausible. Every

numeric parameter is tagged as fixed by design, calibrated against a cited source, or explicitly assumed. A pilot instantiation (Appendix A) shows that the mechanism calibrates to its tier targets and generates the intended confounding, that valid effect estimation is limited by the near-deterministic assignment the targeting figures imply, and that the pre-specified held-out check is informative rather than automatic, since the pilot lands marginally outside its band. The generator notebook and analysis deliverables that implement and exercise this specification are addressed separately; this document defines the structure they are built against.

A Reference Instantiation and Pilot Check

This appendix gives one admissible instantiation of the generative forms that Sections 7–9 leave open and reports what a pilot implementation of it shows. The pilot is a feasibility check of the specification, not the production generator: it implements only the core mechanism (Phases 1–5) and omits the non-core attributes of Appendix B. Every value below is ASSUMED unless marked as calibrated, a script reproducing the results accompanies this paper, and an implementation may depart from the instantiation provided it respects the constraints of Sections 5–9.

A.1 Instantiation

- **Entry attributes and baseline vulnerability (Phases 1–2).** $\text{tier}_i \sim \text{Bernoulli}(0.15)$, $\text{work}_i \sim \text{Bernoulli}(0.35)$, and $\text{inc}_i, \varepsilon_i \sim N(0,1)$ independently, where inc_i is a standardized income proxy whose quantiles define the exported income bracket. Baseline vulnerability v_i is $\delta\,\text{tier}_i + 0.5\,\text{work}_i - 0.4\,\text{inc}_i + 0.8\,\varepsilon_i$, standardized across the cohort to mean 0 and standard deviation 1; δ is calibrated.

- **Term shock.** $\text{shock}_{i,t} \sim \text{Bernoulli}(0.08)$, independent of every other quantity.

- **Observables (Phase 3).** With $e^g, e^a \sim N(0,1)$ independent: $\text{grade}_{i,t} = 72 - 7v_i - 7\,\text{shock}_{i,t} + 8e^g_{i,t}$ and $\text{attendance}_{i,t} = 0.88 - 0.06v_i - 0.08\,\text{shock}_{i,t} + 0.08e^a_{i,t}$ (clipped to $[0,100]$ and $[0,1]$ when exported; clipping does not change the comparisons with τ_g and τ_a), and $\text{hold}_{i,t} \sim \text{Bernoulli}\big(\sigma(-2.2 + 0.9v_i + 1.2\,\text{shock}_{i,t})\big)$. The three are independent given $(v_i, \text{shock}_{i,t})$, as Section 7 requires.

- **Assignment (Phase 4).** Eqs. 1–2 with $\tau_g = 60$, $\tau_a = 0.75$, $\beta_0 = -9$ and $\beta = 6$, and no assignment in term 1.

- **Outcome (Phase 5).** Eq. 3 with $\gamma_1 = 0.8$ and $\gamma_2 = 1.5$ fixed, and γ_0 and δ calibrated, using common random numbers and nested bisection (the typical-tier rate is monotone in γ_0 and the high-tier rate in δ), to the two Class of 2022 tier targets; the result is $\gamma_0 = -2.67$ and $\delta = 1.06$.

A.2 Results

Table 6 summarizes the pilot at $N = 50{,}000$ students and eight terms; Table 7 gives the assignment cells behind the positivity finding.

A.3 Reading the results

Four conclusions follow, each limited by this being one instantiation with illustrative forms. First, the specified mechanism is calibratable: two tier targets are met exactly with one free hazard intercept and one tier shift. Second, the confounding the paper promises is generated at a scale

Table 6: Pilot results for the reference instantiation (seeded, reproducible with the accompanying script)

Check	**Outcome**
Calibration	High tier 67.1%, typical tier 45.9%, pooled 49.1%: both targets are met and the pooled rate equals the derived blend. Across 20 further populations at the calibrated parameters the tier rates are $67.8 \pm 0.6\%$ and $45.7 \pm 0.2\%$.
Confounding presence	Naive $1 - \widehat{RR}$ is -57.6% (95% CI -64.1% to -50.8%, student-level bootstrap), so the interval excludes the true 30% and the benchmark is met.
Inverse-probability weighting	With the true propensity: 75.6% untrimmed (weights up to 8,104), -17.8% trimmed to $[0.05, 0.95]$. Not recovered.
Full-population standardization	73.8%. Not recovered.
Logistic regression	Odds ratio 0.718, an apparent 28.2% reduction: close here, but not the target quantity (Section 11).
Mantel–Haenszel and Poisson	Both give a 25.5% reduction; the Poisson 95% interval for $1 - RR$ is 16.6% to 33.5% and includes 30%. Recovered.
Estimator variability	Over 20 independent populations the Mantel–Haenszel estimate averages 28.4% with a standard deviation of 4.4 points (range 21.0% to 35.4%); 75% of runs fall inside a fixed 25–35% window.
Held-out cohort (Class of 2023)	Refitting only γ_0 (from -2.67 to -3.48) to a pooled 29.4% gives high tier 45.5%, typical tier 26.5%, and a tier ratio of 1.72, against the pre-specified band of 1.2 to 1.7 and an observed ratio of 1.37. Over 8 independent populations the ratio is 1.71 ± 0.04 and 12% of runs fall inside the band. Not passed.

Table 7: Assignment by prior-term signal count k (student-terms with $t \geq 2$, pilot)

k	Student-terms	Treated	Share treated
0	186,580	19	0.01%
1	47,341	2,258	4.8%
2	11,407	10,880	95.4%
3	2,168	2,167	99.95%

that cannot be mistaken for the true effect. Third, the targeting figures of Section 8 force near-deterministic assignment in two of the four cells of Table 7, which is why estimators that need overlap fail and estimators that assume a constant risk ratio succeed. The constant risk ratio is true by construction (Eq. 3), so this is a statement about what the data can identify without that assumption, not about the estimators' general merit. It follows from Eqs. 1–2 and Table 3 and does not depend on the Phase 3 forms. Fourth, the held-out check is not automatic: the single logit-scale level shift predicts a wider tier gap (1.71) than the data showed between the two cohorts (1.37 on the three high-tier regions with a reported Class of 2023 value), so a hazard with this structure is too rigid to reproduce how the gap moved. This is the kind of failure the check exists to expose. The band was fixed before the pilot was run and has not been adjusted, and the result does not show the architecture is wrong, only that this instantiation does not pass.

B Field-Level Schema and Reproducibility Policy

B.1 Schema

Table 8 gives the exported columns of the eight tables of Table 2. The panel tables use the composite key (`student_id`, `term`) and hold one row per student per term for terms $1, \ldots, T_i$ only.

Table 8: Exported columns by table

Table	Columns (type)	Key
Regions	`region_tier` ({high, typical}), `tier_share` (float), `target_attrition` (float), `target_source` (string)	`region_tier`
Students	`student_id` (integer), `region_tier` ({high, typical}), `entry_year` (integer), `program` (categorical), `income_bracket` (ordinal, 1 to 5), `working_student` (boolean), `distance_km` (float $\geq$ 0), `subsidy_at_entry` (boolean)	`student_id`; `region_tier` references Regions
Enrollments	`student_id`, `term` (integer, 1 to 8), `enrollment_status` ({continuing, withdrawn}), `censored` (boolean, true only on the term-8 row of a student who never withdrew)	student, term
CourseGrades	`student_id`, `term`, `term_avg_score` (float, 0 to 100), `units_attempted` (integer $\geq$ 0), `units_passed` (integer, at most `units_attempted`)	student, term
AttendanceRecords	`student_id`, `term`, `attendance_rate` (float, 0 to 1)	student, term
FinancialHolds	`student_id`, `term`, `hold_flag` (boolean), `overdue_amount` (float $\geq$ 0, zero when no hold), `subsidy_status` (boolean)	student, term
SupportInterventions	`student_id`, `term`, `risk_score` (float in $(0, 1)$, null in term 1), `intervention` (boolean, always false in term 1)	student, term
PersistenceOutcomes	`student_id`, `term`, `dropout_flag` (boolean), `attrition_flag` (boolean, true on every row of a student who withdraws within eight terms)	student, term

B.2 Integrity rules

- Never exported: baseline vulnerability, the shock indicators, the hazard $h_{i,t}$, and any calibration intermediate.
- No panel row exists after a student's dropout term, and every `student_id` in a panel table appears in Students.
- Leakage rule: no exported variable may be generated from the dropout outcome of its own or a later term, or from later-term observables. The non-core attributes (`program`, `distance_km`, the unit counts, `overdue_amount`, and the subsidy fields) may depend only on entry attributes, the latent state, and current-term observables; beyond that their marginal forms are unconstrained by this paper.
- The risk score and the intervention are undefined and false, respectively, in a student's first term (Section 8).

B.3 Reproducibility policy

1. Each build uses one integer master seed, recorded in the dataset metadata.
2. Random streams are independent per phase and per variable and derived from the master seed by a documented spawning scheme (for example NumPy's `SeedSequence`), so adding or changing one variable does not alter the draws of any other.
3. Calibration uses common random numbers: the uniform and normal draws are held fixed while parameters vary, so that tier-level attrition is a monotone function of the calibrated parameters.
4. Each release records the specification version, the generator version, the master seed, the library versions, the full tagged parameter log, and a content hash of every exported table.
5. Reproduction is exact only on the same platform and library versions; across platforms, equivalence is statistical, meaning that the checks of Table 5 agree within sampling error.

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